
■ ■ NIDS — AI Network Intrusion Detection Forensic Analysis Report

Session: 2026-09-25 19:32:39 — 2026-09-25 19:45:20 | Host: Binoy | Mode: Simulate

Executive Summary

■■ THREATS DETECTED — Immediate investigation required.

Metric	Value
Total Flows Analysed	500
Clean Flows	0
■ Critical Threats	500
■ High Threats	0
■ Medium Threats	0
■ Low Threats	0
Clean Traffic %	0.0%
Threat Traffic %	100.0%
Session Duration	00:12:40
ML Model Accuracy	Trained (see ml/models/)

Alert Log (100 entries)

Time	Severity	Source	Destination	Attack	Confidence
19:45:20	CRITICAL	192.168.1.29:56543	192.168.1.1:53	EXPLOIT	63.9%
19:45:19	CRITICAL	192.168.1.2:41641	192.168.1.1:22	EXPLOIT	63.9%
19:45:19	CRITICAL	192.168.1.4:23839	192.168.1.1:3306	EXPLOIT	51.3%
19:45:19	CRITICAL	192.168.1.29:39721	192.168.1.1:53	EXPLOIT	63.9%
19:45:18	CRITICAL	192.168.1.4:29361	192.168.1.1:25	EXPLOIT	64.7%
19:45:18	CRITICAL	192.168.1.10:35812	192.168.1.1:25	EXPLOIT	64.4%
19:45:18	CRITICAL	192.168.1.21:4963	192.168.1.1:443	EXPLOIT	63.9%
19:45:18	CRITICAL	192.168.1.11:3604	192.168.1.1:22	EXPLOIT	58.4%
19:45:18	CRITICAL	192.168.1.2:49217	192.168.1.1:22	EXPLOIT	58.4%
19:45:17	CRITICAL	192.168.1.20:54168	192.168.1.1:443	EXPLOIT	63.9%
19:45:17	CRITICAL	192.168.1.6:59539	192.168.1.1:53	EXPLOIT	64.2%
19:45:17	CRITICAL	10.0.222.64:24986	192.168.1.1:23	EXPLOIT	63.9%
19:45:17	CRITICAL	192.168.1.9:34787	192.168.1.1:22	EXPLOIT	50.7%
19:45:17	CRITICAL	192.168.1.20:2423	192.168.1.1:80	EXPLOIT	63.9%
19:45:16	CRITICAL	192.168.1.9:7305	192.168.1.1:80	EXPLOIT	63.9%
19:45:16	CRITICAL	192.168.1.14:22687	192.168.1.1:443	EXPLOIT	64.2%
19:45:16	CRITICAL	192.168.1.23:20509	192.168.1.1:25	EXPLOIT	64.4%
19:45:16	CRITICAL	192.168.1.19:21515	192.168.1.1:3306	EXPLOIT	51.3%
19:45:16	CRITICAL	192.168.1.19:56760	192.168.1.1:80	EXPLOIT	63.9%
19:45:15	CRITICAL	192.168.1.11:40251	192.168.1.1:80	EXPLOIT	63.9%
19:45:15	CRITICAL	192.168.1.13:48465	192.168.1.1:22	EXPLOIT	64.7%
19:45:15	CRITICAL	10.0.222.64:41774	192.168.1.1:23	EXPLOIT	64.9%
19:45:15	CRITICAL	192.168.1.28:11494	192.168.1.1:3306	EXPLOIT	50.0%
19:45:14	CRITICAL	192.168.1.9:37874	192.168.1.1:53	EXPLOIT	63.9%
19:45:14	CRITICAL	192.168.1.16:2342	192.168.1.1:3306	EXPLOIT	50.0%
19:45:14	CRITICAL	192.168.1.2:3353	192.168.1.1:22	EXPLOIT	64.4%
19:45:13	CRITICAL	192.168.1.17:38004	192.168.1.1:53	EXPLOIT	63.9%
19:45:13	CRITICAL	192.168.1.7:51837	192.168.1.1:25	EXPLOIT	58.9%
19:45:13	CRITICAL	192.168.1.12:51083	192.168.1.1:3306	EXPLOIT	51.0%
19:45:12	CRITICAL	192.168.1.26:11433	192.168.1.1:53	EXPLOIT	63.9%
19:45:12	CRITICAL	192.168.1.29:57621	192.168.1.1:25	EXPLOIT	47.4%
19:45:11	CRITICAL	192.168.1.3:54316	192.168.1.1:443	EXPLOIT	63.4%
19:45:11	CRITICAL	192.168.1.10:10861	192.168.1.1:3306	EXPLOIT	51.0%
19:45:11	CRITICAL	192.168.1.12:8617	192.168.1.1:3306	EXPLOIT	51.3%
19:45:10	CRITICAL	192.168.1.18:31883	192.168.1.1:80	EXPLOIT	63.9%
19:45:10	CRITICAL	192.168.1.16:54762	192.168.1.1:443	EXPLOIT	63.9%
19:45:10	CRITICAL	192.168.1.22:46411	192.168.1.1:53	EXPLOIT	63.9%
19:45:09	CRITICAL	10.0.222.64:13205	192.168.1.1:23	EXPLOIT	63.9%

19:45:09	CRITICAL	192.168.1.5:29937	192.168.1.1:25	EXPLOIT	63.9%
19:45:09	CRITICAL	192.168.1.3:65329	192.168.1.1:25	EXPLOIT	64.7%
19:45:08	CRITICAL	192.168.1.12:8682	192.168.1.1:25	EXPLOIT	64.4%
19:45:08	CRITICAL	192.168.1.24:45095	192.168.1.1:53	EXPLOIT	64.2%
19:45:08	CRITICAL	192.168.1.28:63698	192.168.1.1:22	EXPLOIT	64.7%
19:45:07	CRITICAL	192.168.1.20:46971	192.168.1.1:3306	EXPLOIT	51.0%
19:45:07	CRITICAL	192.168.1.7:7793	192.168.1.1:22	EXPLOIT	64.7%
19:45:07	CRITICAL	192.168.1.29:45497	192.168.1.1:53	EXPLOIT	61.2%
19:45:06	CRITICAL	192.168.1.29:36939	192.168.1.1:3306	EXPLOIT	51.0%
19:45:06	CRITICAL	192.168.1.24:48113	192.168.1.1:22	EXPLOIT	64.7%
19:45:06	CRITICAL	10.0.222.64:27278	192.168.1.1:23	EXPLOIT	63.9%
19:45:05	CRITICAL	192.168.1.19:10982	192.168.1.1:443	EXPLOIT	63.9%
19:45:05	CRITICAL	192.168.1.5:59606	192.168.1.1:22	EXPLOIT	64.4%
19:45:04	CRITICAL	192.168.1.21:61933	192.168.1.1:25	EXPLOIT	64.7%
19:45:04	CRITICAL	192.168.1.26:23367	192.168.1.1:22	EXPLOIT	64.4%
19:45:04	CRITICAL	192.168.1.9:19548	192.168.1.1:3306	EXPLOIT	50.0%
19:45:03	CRITICAL	192.168.1.6:26242	192.168.1.1:22	EXPLOIT	64.4%
19:45:03	CRITICAL	192.168.1.8:53687	192.168.1.1:53	EXPLOIT	63.9%
19:45:03	CRITICAL	192.168.1.25:33240	192.168.1.1:80	EXPLOIT	63.7%
19:45:02	CRITICAL	192.168.1.28:11310	192.168.1.1:80	EXPLOIT	63.9%
19:45:02	CRITICAL	10.0.222.64:63430	192.168.1.1:23	EXPLOIT	64.9%
19:45:02	CRITICAL	192.168.1.2:23217	192.168.1.1:53	EXPLOIT	63.9%
19:45:01	CRITICAL	192.168.1.27:47914	192.168.1.1:53	EXPLOIT	63.9%
19:45:01	CRITICAL	192.168.1.6:51344	192.168.1.1:443	EXPLOIT	63.9%
19:45:01	CRITICAL	192.168.1.29:47338	192.168.1.1:53	EXPLOIT	63.9%
19:45:00	CRITICAL	192.168.1.25:51523	192.168.1.1:53	EXPLOIT	64.2%
19:45:00	CRITICAL	192.168.1.14:5044	192.168.1.1:25	EXPLOIT	64.7%
19:45:00	CRITICAL	192.168.1.12:13994	192.168.1.1:3306	EXPLOIT	52.0%
19:44:59	CRITICAL	192.168.1.27:45698	192.168.1.1:443	EXPLOIT	63.4%
19:44:59	CRITICAL	10.0.222.64:35781	192.168.1.1:23	EXPLOIT	64.9%
19:44:58	CRITICAL	192.168.1.10:26557	192.168.1.1:53	EXPLOIT	61.4%
19:44:58	CRITICAL	192.168.1.11:11011	192.168.1.1:22	EXPLOIT	64.7%
19:44:58	CRITICAL	192.168.1.10:60011	192.168.1.1:80	EXPLOIT	64.2%
19:44:57	CRITICAL	192.168.1.25:6240	192.168.1.1:443	EXPLOIT	58.4%
19:44:57	CRITICAL	192.168.1.8:19904	192.168.1.1:53	EXPLOIT	64.2%
19:44:56	CRITICAL	192.168.1.25:57828	192.168.1.1:3306	EXPLOIT	51.3%
19:44:56	CRITICAL	192.168.1.8:1473	192.168.1.1:22	EXPLOIT	64.7%
19:44:56	CRITICAL	192.168.1.27:55126	192.168.1.1:25	EXPLOIT	64.4%
19:44:55	CRITICAL	10.0.222.64:39986	192.168.1.1:23	EXPLOIT	64.9%
19:44:55	CRITICAL	192.168.1.5:2162	192.168.1.1:3306	EXPLOIT	51.0%
19:44:55	CRITICAL	192.168.1.13:52164	192.168.1.1:443	EXPLOIT	64.2%

19:44:54	CRITICAL	192.168.1.6:36617	192.168.1.1:443	EXPLOIT	63.9%
19:44:54	CRITICAL	192.168.1.10:15711	192.168.1.1:22	EXPLOIT	64.4%
19:44:54	CRITICAL	192.168.1.7:46289	192.168.1.1:443	EXPLOIT	63.9%
19:44:53	CRITICAL	192.168.1.18:59531	192.168.1.1:25	EXPLOIT	64.7%
19:44:53	CRITICAL	192.168.1.20:57455	192.168.1.1:53	EXPLOIT	63.9%
19:44:52	CRITICAL	10.0.222.64:20302	192.168.1.1:23	EXPLOIT	64.4%
19:44:52	CRITICAL	192.168.1.20:36069	192.168.1.1:443	EXPLOIT	64.2%
19:44:52	CRITICAL	192.168.1.16:52594	192.168.1.1:3306	EXPLOIT	51.0%
19:44:51	CRITICAL	192.168.1.28:23363	192.168.1.1:80	EXPLOIT	64.2%
19:44:51	CRITICAL	192.168.1.7:2997	192.168.1.1:25	EXPLOIT	64.7%
19:44:51	CRITICAL	192.168.1.13:17607	192.168.1.1:443	EXPLOIT	63.9%
19:44:50	CRITICAL	192.168.1.29:10612	192.168.1.1:3306	EXPLOIT	51.3%
19:44:50	CRITICAL	192.168.1.28:59967	192.168.1.1:3306	EXPLOIT	50.5%
19:44:50	CRITICAL	192.168.1.4:62584	192.168.1.1:25	EXPLOIT	63.9%
19:44:49	CRITICAL	192.168.1.28:19795	192.168.1.1:53	EXPLOIT	57.9%
19:44:49	CRITICAL	192.168.1.8:17061	192.168.1.1:443	EXPLOIT	64.2%
19:44:49	CRITICAL	192.168.1.12:42163	192.168.1.1:53	EXPLOIT	57.9%
19:44:48	CRITICAL	192.168.1.19:15920	192.168.1.1:80	EXPLOIT	57.9%
19:44:48	CRITICAL	192.168.1.12:2433	192.168.1.1:22	EXPLOIT	64.7%
19:44:48	CRITICAL	192.168.1.5:63979	192.168.1.1:25	EXPLOIT	57.9%
19:44:47	CRITICAL	192.168.1.29:42948	192.168.1.1:3306	EXPLOIT	52.0%

Top Attacker IPs

Rank	IP Address	Alert Count	Threat Level
1	10.0.77.132	1181	CRITICAL
2	192.168.1.25	29	CRITICAL
3	192.168.1.28	28	CRITICAL
4	192.168.1.24	27	CRITICAL
5	192.168.1.20	25	CRITICAL

MITRE ATT&CK; Framework Mapping

Each detected attack class is mapped to its corresponding MITRE ATT&CK; Tactics and Techniques (Framework v14). This enables security teams to correlate NIDS detections with the broader threat intelligence ecosystem.

Attack Class	Tactic	Technique ID	Technique Name	Severity
DOS	Impact	T1498	Network Denial of Service	CRITICAL
DOS	Impact	T1499	Endpoint Denial of Service	CRITICAL
DDOS	Impact	T1498	Network Denial of Service	CRITICAL
DDOS	Impact	T1499	Endpoint Denial of Service	CRITICAL
PORT_SCAN	Discovery	T1046	Network Service Discovery	MEDIUM
PORT_SCAN	Reconnaissance	T1595	Active Scanning	MEDIUM
PORT_SCAN	Reconnaissance	T1590	Gather Victim Network Information	MEDIUM
VULN_SCAN	Reconnaissance	T1595	Active Scanning	HIGH
VULN_SCAN	Reconnaissance	T1590	Gather Victim Network Information	HIGH
VULN_SCAN	Discovery	T1046	Network Service Discovery	HIGH
BRUTE_FORCE	Credential Access	T1110	Brute Force	HIGH
BRUTE_FORCE	Initial Access	T1021	Remote Services	HIGH
BRUTE_FORCE	Initial Access	T1078	Valid Accounts	HIGH
EXPLOIT	Privilege Escalation	T1068	Exploitation for Privilege Escalation	CRITICAL
EXPLOIT	Privilege Escalation	T1055	Process Injection	CRITICAL
EXPLOIT	Privilege Escalation	T1059	Command and Scripting Interpreter	CRITICAL
EXPLOIT	Privilege Escalation	T1548	Abuse Elevation Control Mechanism	CRITICAL
WEB_ATTACK	Initial Access	T1190	Exploit Public-Facing Application	HIGH
WEB_ATTACK	Privilege Escalation	T1059	Command and Scripting Interpreter	HIGH
WEB_ATTACK	Lateral Movement	T1210	Exploitation of Remote Services	HIGH
INFILTRATION	Command and Control	T1071	Application Layer Protocol	CRITICAL
INFILTRATION	Command and Control	T1105	Ingress Tool Transfer	CRITICAL
INFILTRATION	Initial Access	T1021	Remote Services	CRITICAL
INFILTRATION	Initial Access	T1133	External Remote Services	CRITICAL
EXFILTRATION	Exfiltration	T1041	Exfiltration Over C2 Channel	CRITICAL
EXFILTRATION	Exfiltration	T1048	Exfiltration Over Alternative Protocol	CRITICAL
EXFILTRATION	Exfiltration	T1567	Exfiltration Over Web Service	CRITICAL

Full details at: <https://attack.mitre.org>

Attack Class Summaries

DOS — Impact (TA0040)

Single-source DoS floods targeting network bandwidth or service availability. Detected via packet_rate, error_rate and count anomalies.

ID	Technique	Description (excerpt)
T1498	Network Denial of Service	Adversaries flood network infrastructure to degrade or block availability for target users...
T1499	Endpoint Denial of Service	Adversaries perform DoS attacks to degrade or block availability of targeted resources at ...

DDOS — Impact (TA0040)

Distributed multi-source flooding — significantly higher packet volume than single-source DoS. Amplification and reflection techniques are common.

ID	Technique	Description (excerpt)
T1498	Network Denial of Service	Adversaries flood network infrastructure to degrade or block availability for target users...
T1499	Endpoint Denial of Service	Adversaries perform DoS attacks to degrade or block availability of targeted resources at ...

PORT_SCAN — Discovery (TA0007) + Reconnaissance (TA0043)

Sequential port probing using Nmap, ipsweep, or portsweep to discover open services. Detected via high diff_srv_rate and RST ratio with low bytes.

ID	Technique	Description (excerpt)
T1046	Network Service Discovery	Adversaries scan victim networks to discover services running on remote hosts. Tools like ...
T1595	Active Scanning	Adversaries actively probe victim infrastructure using port scans, vulnerability scans, an...
T1590	Gather Victim Network Information	Adversaries gather information about the target's network infrastructure including IP rang...

VULN_SCAN — Reconnaissance (TA0043)

Targeted vulnerability enumeration against specific services using tools like Satan, Nessus, or OpenVAS. More selective than port scanning.

ID	Technique	Description (excerpt)
T1595	Active Scanning	Adversaries actively probe victim infrastructure using port scans, vulnerability scans, an...
T1590	Gather Victim Network Information	Adversaries gather information about the target's network infrastructure including IP rang...
T1046	Network Service Discovery	Adversaries scan victim networks to discover services running on remote hosts. Tools like ...

BRUTE_FORCE — Credential Access (TA0006)

Repeated authentication attempts against SSH, FTP, or HTTP. Detected via high same_srv_rate, error_rate, and repeated connections to auth ports 21/22/80.

ID	Technique	Description (excerpt)
T1110	Brute Force	Adversaries use brute force techniques to attempt access to accounts when passwords are un...
T1021	Remote Services	Adversaries use valid credentials to log into a service specifically designed to accept re...
T1078	Valid Accounts	Adversaries obtain and abuse credentials of existing accounts as a means of gaining initia...

EXPLOIT — Privilege Escalation (TA0004)

Buffer overflow, shellcode injection, or kernel module exploitation to gain root/SYSTEM access. Detected via very high src_bytes, long duration, and urgent flags.

ID	Technique	Description (excerpt)
T1068	Exploitation for Privilege Escalation	Adversaries exploit software vulnerabilities in an attempt to elevate privileges. This is ...
T1055	Process Injection	Adversaries inject code into processes to evade process-based defences and possibly elevat...
T1059	Command and Scripting Interpreter	Adversaries abuse command and script interpreters to execute commands, scripts, or binarie...
T1548	Abuse Elevation Control Mechanism	Adversaries bypass mechanisms designed to control elevate privileges to perform privileged...

WEB_ATTACK — Initial Access (TA0001) + Execution (TA0002)

SQL injection, XSS, or HTTP exploitation targeting web applications on port 80/443. Detected via crafted payloads in HTTP requests with unusual byte patterns.

ID	Technique	Description (excerpt)
T1190	Exploit Public-Facing Application	Adversaries exploit weaknesses in internet-facing software including web applications, dat...
T1059	Command and Scripting Interpreter	Adversaries abuse command and script interpreters to execute commands, scripts, or binarie...
T1210	Exploitation of Remote Services	Adversaries exploit remote services to gain access to internal systems. This includes expl...

INFILTRATION — Command and Control (TA0011) + Lateral Movement (TA0008)

Persistent backdoor or C2 channel maintaining covert access. Detected via long session duration on non-standard ports with low but steady traffic patterns.

ID	Technique	Description (excerpt)
T1071	Application Layer Protocol	Adversaries communicate using application layer protocols to avoid detection. HTTP, HTTPS,...
T1105	Ingress Tool Transfer	Adversaries transfer tools or malware from external systems into a compromised network env...
T1021	Remote Services	Adversaries use valid credentials to log into a service specifically designed to accept re...

T1133	External Remote Services	Adversaries leverage external-facing remote services such as VPN, Citrix, RDP, and SSH to ...
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EXFILTRATION — Exfiltration (TA0010)

Large unauthorised outbound data transfer via FTP, HTTP, or covert channels. Detected via abnormally high dst_bytes and byte_rate on standard ports.

ID	Technique	Description (excerpt)
T1041	Exfiltration Over C2 Channel	Adversaries steal data over the existing C2 channel. This avoids creating new outbound con...
T1048	Exfiltration Over Alternative Protocol	Adversaries use protocols other than the C2 channel to exfiltrate data — DNS tunnelling, I...
T1567	Exfiltration Over Web Service	Adversaries use legitimate web services (Google Drive, Dropbox, GitHub, Pastebin) to exfil...

AI Explanation Samples (Top 3 Alerts)

Alert: EXPLOIT from 192.168.1.29:56543 — Confidence: 63.9%

Feature	Value	Contribution	Direction
wrong_fragment	0.0	38.8%	↓ decreases risk
srv_count	1.0	23.9%	↑ increases risk
count	1.0	10.1%	↑ increases risk
src_bytes	1163.0	10.0%	↑ increases risk
urgent	0.0	7.6%	↓ decreases risk

Alert: EXPLOIT from 192.168.1.2:41641 — Confidence: 63.9%

Feature	Value	Contribution	Direction
wrong_fragment	0.0	36.9%	↓ decreases risk
srv_count	1.0	23.1%	↑ increases risk
count	1.0	9.8%	↑ increases risk
src_bytes	117.0	9.7%	↑ increases risk
urgent	0.0	7.8%	↓ decreases risk

Alert: EXPLOIT from 192.168.1.4:23839 — Confidence: 51.3%

Feature	Value	Contribution	Direction
wrong_fragment	0.0	34.4%	↓ decreases risk
srv_count	1.0	22.8%	↑ increases risk
urgent	0.0	14.4%	↓ decreases risk
count	1.0	12.0%	↑ increases risk
src_bytes	436.0	9.1%	↑ increases risk

Recommendations

1. Block attacker IPs

Use firewall rules to block the top source IPs identified in the alert log.

2. Investigate DoS flows

High packet-rate flows from single sources should be traced to origin. Consider rate limiting at the network edge.

3. Review probe targets

Port scan targets indicate which services are being enumerated. Disable unused services and apply strict firewall ingress rules.

4. Patch R2L vulnerabilities

Remote-to-Local attack attempts suggest exposed services (SSH, FTP, SMTP). Enforce key-based auth and disable password login.

5. Enable continuous monitoring

Deploy NIDS in persistent mode with SIEM integration and automated alerting for 24/7 coverage.

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